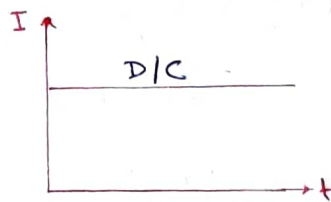


Alternating Current

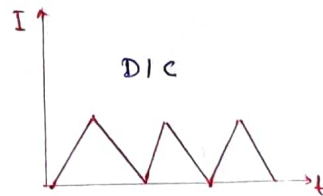
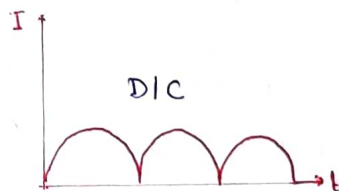
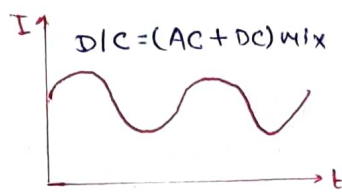
Direct current (D/C):-

- Unidirectional (steady current)
- Magnitude may be variable or constant.

constⁿ D/C

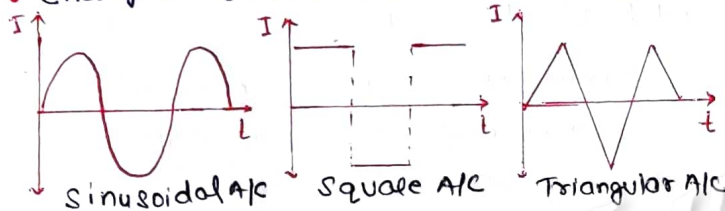


variable D/C



Alternating current:-

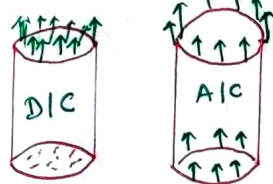
- Bidirectional
- Changes its direction periodically.



⇒ Disadvantages of D/C :-

- ① D/C can't be supply over large distance but A/C can be supply over large dist.
- ② Power loss is max in transmission but voltage of A/C can be step-up or down by using transformers.
- ③ D/C supply at constant voltage.

$I_{D/C} \propto r^{3/2}$
 $I_{A/C} \propto \text{surface area}$



⇒ Disadvantage of A/C:-

- A/C can't be store.
- Skin effect and corona discharge.

⇒ Variable D/C or A/C

Discrete variation

Continuous variable

$$I_{avg} = I_{mean} = \frac{i_1 + i_2 + \dots + i_n}{n}$$

$$\langle i \rangle = i_{avg} = \frac{\int i \cdot dt}{\int dt}$$

R.M.S value

Root ← Mean ← Square value

Reading of voltmeter = Effect = virtual current
 or Ammeter current

① R.M.S current

$$I_{rms} = \sqrt{\langle i^2 \rangle} = \sqrt{\frac{\int i^2 \cdot dt}{\int dt}}$$

② R.M.S voltage

$$V_{rms} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{\int v^2 \cdot dt}{\int dt}}$$

③ Heat dissipated

$$\langle \text{Energy} \rangle = \langle I^2 \rangle R t = I_{rms}^2 R t$$

MR RATTA*

Full cycle (Fc)

Half cycle (Hc)

$$\langle \sin \theta \rangle = 0$$

$$\langle \sin \theta \rangle = \frac{2}{\pi}$$

$$\langle \cos \theta \rangle = 0$$

$$\langle \cos \theta \rangle = \frac{2}{\pi}$$

$$\langle \sin^2 \theta \rangle = \langle \cos^2 \theta \rangle = \frac{1}{2} \quad \text{Half / Full cycle}$$

⇒ for Sinusoidal AC

$$[I = I_0 \sin \omega t]$$

$$[V = V_0 \sin \omega t]$$

$$\langle I \rangle_{fc} = 0$$

$$\langle V \rangle_{fc} = 0$$

$$\langle I \rangle_{hc} = \frac{2I_0}{\pi}$$

$$\langle V \rangle_{hc} = \frac{2V_0}{\pi}$$

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

$$V_{rms} = \frac{V_0}{\sqrt{2}}$$

Alternating current

$$I = I_0 \sin(\omega t + \phi)$$

I_0 = Max. current ω = angular frequency = $2\pi f$

I = Instant. current ϕ = phase difference.

Moving coil galvanometer

Hot-wire Ammeter

- Torque on current carrying coil.
- only D/C can measure
- $\theta \propto I$
- Linear scale.

- Heat loss
- A/C / D/C both can be measure.
- $H \propto \theta \propto I^2$
- Non-linear scale.

MR* Table

Current	Half Avg	Full Avg	RMS
$I = I_0 \sin(\omega t)$ (sinusoidal)	$\frac{2I_0}{\pi}$	0	$\frac{I_0}{\sqrt{2}}$
Square wave	I_0	0	I_0
Triangular AC	$I_0/2$	0	$\frac{I_0}{\sqrt{3}}$
Full wave Rectified	$\frac{2I_0}{\pi}$	$\frac{2I_0}{\pi}$	$\frac{I_0}{\sqrt{2}}$
Half wave Rectified	$\frac{2I_0}{\pi}$	$\frac{I_0}{\pi}$	$\frac{I_0}{2}$

MR* DC AC

(i) $i = a + b \sin(\omega t)$

$$i_{rms} = \sqrt{\frac{a^2 + \frac{b^2}{2}}{2}}$$

(ii) $i = a \sin \omega t + b \cos \omega t$

$$i_{rms} = \sqrt{\frac{a^2 + b^2}{2}}$$

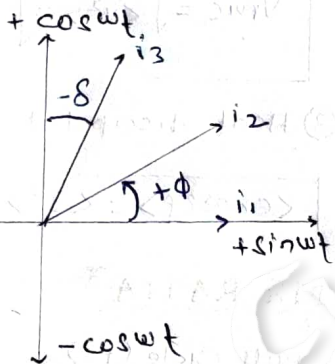
Phasor diagram

$$i_1 = i_0 \sin \omega t$$

$$i_2 = i_0 \sin(\omega t + \phi)$$

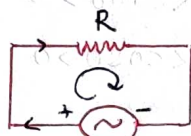
$$i_3 = i_0 \cos(\omega t - \delta)$$

$$-\sin \omega t$$



AC Source across pure Resistance

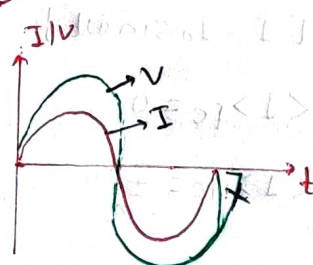
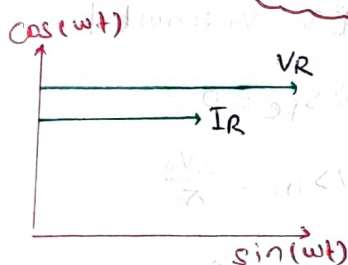
$I_R = I_0 \sin(\omega t)$
 $V_R = E_R = E_0 \sin(\omega t)$ In same phase



$$I_0 = \frac{E_0}{R}$$

$\langle P \rangle = I_{rms} V_{rms}$
 $\langle P \rangle = \frac{I_0 V_0}{2}$

$\phi = 0$

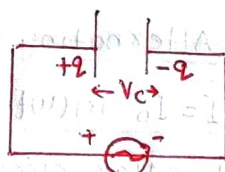


AC Source across pure Capacitor

$$E = E_0 \sin \omega t$$

$$I = I_0 \cos \omega t$$

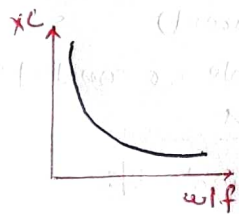
'I' leads 'V'
by $\pi/2$ i.e. $\pi/4$



$$I = \frac{E_0}{1/\omega C} \cos \omega t = \frac{E_0}{X_C} \cos \omega t$$

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

capacitive Reactance



* capacitor is DC filter circuit.

for DC

$$f = 0$$

$$X_C = \infty$$

capacitor act as open wire

for AC

$$f = \omega$$

$$X_C = 0$$

capacitor act as simple wire

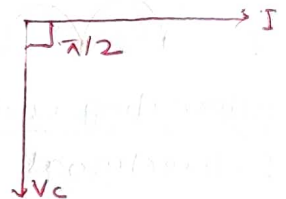
⇒ Power loss across capacitor

$$\cos \phi = 0$$

$$\phi = \frac{\pi}{2}$$

$$\langle P \rangle = 0$$

wattless current

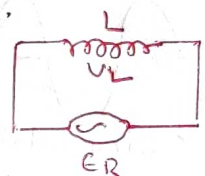


AC source across pure Inductor

$$V_L = E_0 \sin(\omega t)$$

$$I_L = I_0 \sin(\omega t - \pi/2)$$

by $\pi/2$

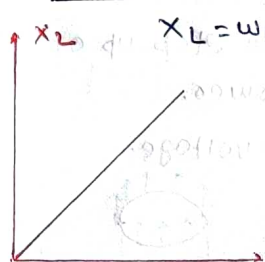


$X_L = \omega L \rightarrow$ Inductive Reactance

$$I_0 = \frac{E_0}{X_L}$$

⇒ Inductive Reactance

$$X_L = \omega L = 2\pi f L$$



for DC

$$f = 0$$

$$X_L = 0$$

$$I \neq 0$$

Simple wire

for AC

$$f = \omega$$

$$X_L = \infty$$

$$I = 0$$

Open wire

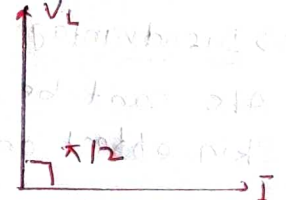
power loss across Inductor

$$\cos \phi = 0$$

$$\phi = \frac{\pi}{2}$$

$$\langle P \rangle = 0$$

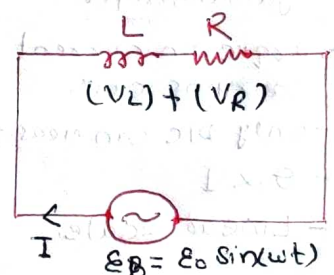
wattless current



Series R-L circuit

$$E_{net} = E_R = E_0 \sin \omega t$$

$$I = I_0 \sin(\omega t - \phi)$$

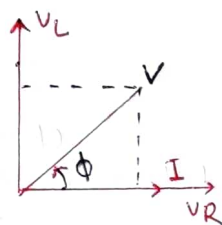


$$V = \sqrt{V_L^2 + V_R^2} \quad V_{\text{net}} \text{ lead 'i' by } \phi$$

$$V_R = iR, V_L = iX_L$$

$$Z = \sqrt{R^2 + X_L^2} \quad \text{Impedance}$$

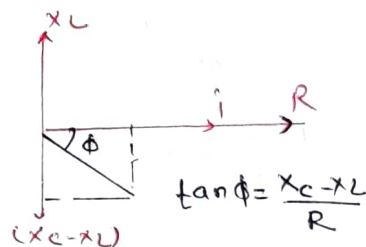
$$\tan \phi = \frac{V_L}{V_R} = \frac{X_L}{R} \quad \cos \phi = \frac{R}{Z} \quad \text{Power factor}$$



Case I:- $X_C > X_L$ (or) $V_C > V_L$

- Capacitive dominance
- Current will lead by voltage.

$$Z = \sqrt{(X_C - X_L)^2 + R^2}$$



Case II:- $X_L = X_C$ (or) $V_L = V_C$

- Resonance condition
- Current max.
- Impedance min.

$$\phi = 0 \quad Z = R \quad \cos \phi = 1$$

$$\omega L = \frac{1}{\omega C} \Rightarrow \omega = \frac{1}{\sqrt{LC}} \Rightarrow f = \frac{1}{2\pi\sqrt{LC}}$$

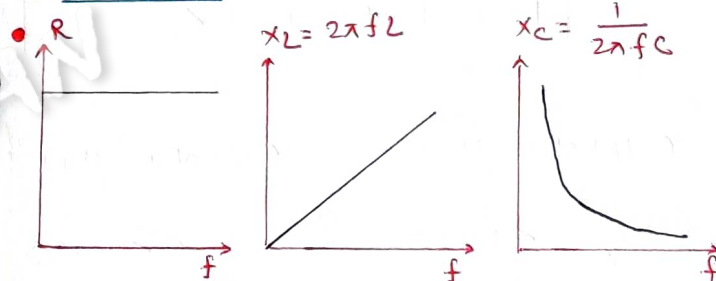
Power drop in series LCR circuit

$$\langle P \rangle = I_{\text{rms}} V_{\text{rms}} \cos \phi \quad \because \cos \phi = \frac{R}{Z}$$

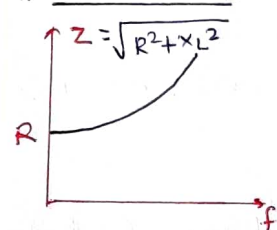
$$\langle P \rangle = I_{\text{rms}}^2 V_{\text{rms}} \frac{R}{Z}$$

$$\langle P \rangle = \frac{V_{\text{rms}}^2 \times R}{Z^2} \quad \langle P \rangle = \frac{V_{\text{rms}}^2 R}{Z^2}$$

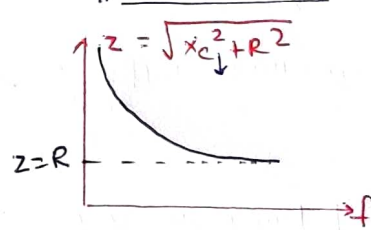
Graph:-



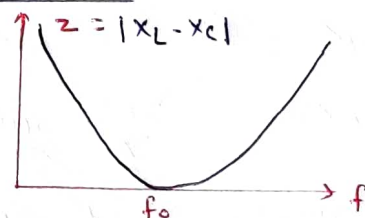
R-L circuit



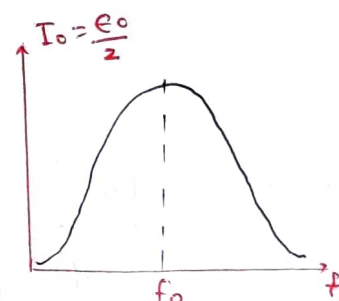
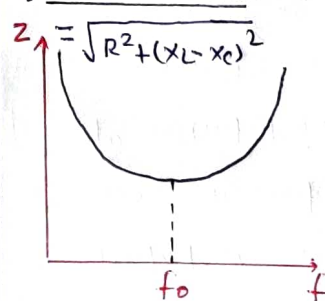
R-C circuit



L-C circuit



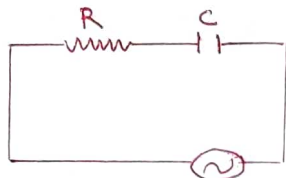
L-C-R circuit



Series RC circuit

$$I = I_0 \sin(\omega t + \phi)$$

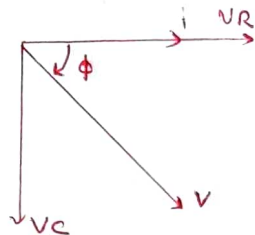
$$I_0 = \frac{\epsilon_0}{Z}$$



$$V = \sqrt{V_R^2 + V_C^2} \quad \text{'i' lead 'V' by } \phi$$

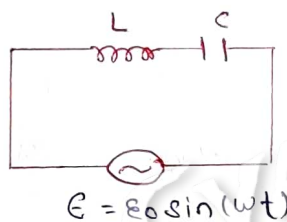
$$Z = \sqrt{R^2 + X_C^2} \quad \tan \phi = \frac{X_C}{R}$$

$$\cos \phi = \frac{R}{Z}$$



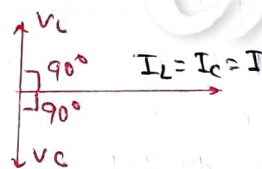
Series L-C circuit

$$V_L - V_C = V_{\text{net}} \quad \phi$$



$$I_0 = \frac{\epsilon_0}{Z}$$

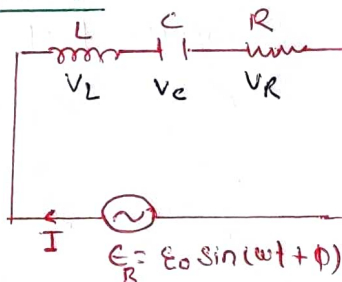
$$Z = X_L - X_C$$



Series L-C-R circuit

$$I = I_R = I_L = I_C \quad (\text{due to series combin.})$$

$$\epsilon_{\text{st}} = \epsilon_{\text{Lt}} + \epsilon_{\text{Ct}} + \epsilon_{\text{Rt}} \quad (\text{at any time 't'})$$



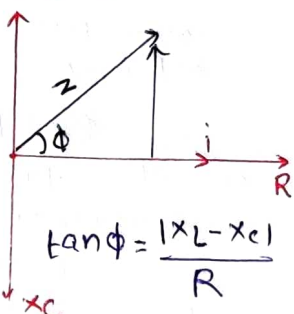
Impedance triangle for L-C-R circuit

Case I:- $X_L > X_C$ (or) $V_L > V_C$

- Inductive dominance ($X_L - X_C$)
- Voltage will lead by current.

$$Z = \sqrt{(X_L - X_C)^2 + R^2}$$

$$\cos \phi = \frac{R}{Z}$$



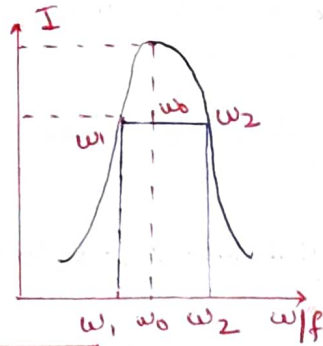
Quality factor

- Q-factor & sharpness of resonance

Band width

$$\omega_2 - \omega_1 = \Delta\omega = \frac{R}{L}$$

$$(\omega_2 - \omega_0) = (\omega_0 - \omega_1) = \frac{\Delta\omega}{2} = \frac{R}{2L}$$



$$Q = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0 L}{R} = \left(\frac{X_L}{R}\right) = \left(\frac{V_L}{R}\right)$$

$$Q = \frac{\omega_0}{\Delta\omega} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Choke coil

- Control current in AC circuit
- Divide Potential without power loss

(X_L, R) choke coil → RL circuit

Ideal

$$\begin{aligned} R &= 0 \\ X_L &\neq 0 \\ \phi &= 90^\circ \\ \cos\phi &= 0 \end{aligned}$$

Practical

$$\begin{aligned} R &= \text{low} \\ X_L &= \text{High} \\ \phi &< 90^\circ \\ \cos\phi &= \text{v. low} \end{aligned}$$

Choke

$$\begin{aligned} R &= \text{high} \\ X_L &= \text{low} \\ \phi &= \text{low} \\ \cos\phi &= \text{high} \end{aligned}$$

Transformer

- Principle:- Mutual inductance.
- use as a voltage regulator.
- only use for AC current/voltage

for Ideal transformer

$$P_{out} = P_{in} \quad \leftarrow \quad \eta = 100\% \quad \leftarrow \quad \eta = \frac{P_{out}}{P_{in}}$$

Efficiency = 100%
Power loss = 0

→ Power doesn't depend on no. of turns
voltage ∝ no. of turns

$$I \propto \frac{1}{\text{no. of turns}}$$

$$\frac{I_{out}}{I_{in}} = \frac{V_{in}}{V_{out}} = \frac{N_p}{N_s}$$

Two types → Step up
Step down

Step-up

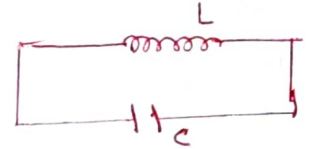
- voltage Increases
- $V_{out} > V_{in}$
- $N_s > N_p$
- $I_{out} < I_{in}$
- $P_{in} = P_{out}$

Step-down

- voltage decreases
- $V_{out} < V_{in}$
- $N_s < N_p$
- $I_{out} > I_{in}$
- $P_{in} = P_{out}$

L-C oscillation

$$\text{At } t=0 \quad U_C = \frac{Q_0^2}{2C}, U_B = 0$$



$$\text{At } t = T/4$$

$$U_C = 0, U_B = \frac{1}{2} Li^2$$

* Energy stored in capacitor in form of $\vec{E}$.

$$\text{At } t = T/2$$

$$U_C = \frac{Q_0^2}{2C}, U_B = 0$$

* Energy stored in inductor in the form of $\vec{B}$.

$$\text{At } t = 3T/4$$

$$U_C = 0, U_B = \frac{1}{2} Li^2$$

$$\text{At } t = T$$

$$U_C = \frac{Q_0^2}{2C}, U_B = 0$$

$$T = \frac{1}{f} = 2\pi\sqrt{LC}$$

Spring Mass

LC oscillation

Mechanical system	Electrical system
Mass (M)	Inductance (L)
Position (x)	charge (q)
Spring const (K)	capacitance (1/C)
velocity (v)	current (I)
$\frac{1}{2} Mv^2$	$\frac{1}{2} Li^2$
$\frac{1}{2} Kx^2$	$\frac{1}{2} \frac{Q^2}{C}$